

HTML

1. Create a basic HTML webpage using **headings, paragraphs, and line breaks**.
2. Create an HTML page containing **ordered, unordered**, and description lists.
3. Create a webpage using **tables** with rows, columns, borders, and merged cells.
4. Create an HTML page containing images and **hyperlinks**.
5. Create a student registration **form** using HTML form elements.
6. Create a webpage using **audio** and **video** elements.
7. Create a webpage using frames/iframes.
8. Create an HTML page demonstrating semantic elements such as <header>, <nav>, <section>, <article>, and <footer>.
9. Create a webpage using HTML5 **input** types such as email, date, number, and password.
10. Create a college timetable using an HTML **table**.
11. Create a webpage with **internal** and **external CSS** styling.
12. Create a webpage demonstrating CSS **selectors, colors, fonts, margins, padding, and borders**.
13. Create a responsive webpage using **CSS media queries**.
14. Create a webpage using Bootstrap components.
15. Create a simple personal portfolio webpage using HTML and CSS.

CSS

CSS Foundation & Core UI

1. Create a webpage using Inline, Internal, and External CSS.
2. Create a webpage using Universal, Element, Class, ID, Attribute, and Grouping selectors.
3. Create a webpage using Descendant, Child, Adjacent Sibling, and General Sibling selectors.
4. Create a webpage using CSS colors for text, backgrounds, borders, and UI elements.
5. Create a webpage using background images, background colors, background positioning, sizing, repeat, and attachment.
6. Create a webpage using different CSS border styles, widths, colors, and border-radius properties.
7. Create a webpage using Linear and Radial CSS gradients.
8. Create a webpage using CSS font-family, font-size, font-weight, font-style, and font-variant properties.
9. Create a webpage using CSS text alignment, decoration, transformation, indentation, spacing, and line-height.
10. Create a webpage using the CSS box model with content, padding, border, and margin.
11. Create a webpage using different width, height, min-width, max-width, min-height, and max-height properties.
12. Create a webpage using the CSS box-sizing property with content-box and border-box.
13. Create a webpage using CSS display properties such as block, inline, inline-block, flex, grid, and none.
14. Create a webpage using CSS visibility and opacity properties.

15. Create a webpage using CSS overflow, overflow-x, and overflow-y properties.
16. Create a webpage using border-radius to create rounded cards, circles, pills, and custom shapes.
17. Create a webpage using CSS box-shadow and text-shadow properties.
18. Create a webpage using CSS variables and custom properties.
19. Create a webpage using CSS inheritance, specificity, and the cascade.
20. Create a webpage using CSS calc(), min(), max(), and clamp() functions.
21. Create a webpage using static, relative, absolute, fixed, and sticky positioning.
22. Create a webpage using z-index and CSS stacking order.
23. Create a webpage using float and clear properties.
24. Create a webpage using different CSS cursor styles and pointer interactions.
25. Create a webpage using CSS outline and outline-offset properties.
26. Create a webpage using CSS margin, padding, and spacing utilities.
27. Create a webpage using CSS width and height percentages, viewport units, and rem units.
28. Create a webpage using CSS em, rem, vw, vh, and percentage units.
29. Create a webpage using CSS object-fit and object-position properties.
30. Create a webpage using CSS filters such as blur, grayscale, brightness, contrast, and opacity.
31. Create a webpage using CSS transitions for colors, sizes, shadows, and positions.
32. Create a webpage using CSS 2D transformations such as rotate, scale, translate, and skew.
33. Create a webpage using CSS 3D transformations and perspective effects.
34. Create a webpage using CSS keyframe animations.
35. Create a webpage using a CSS loading spinner and loading animation.

SaaS UI Components

36. Create a webpage for a SaaS navigation bar.
37. Create a webpage for a SaaS sidebar navigation.
38. Create a webpage for a responsive SaaS navigation menu.
39. Create a webpage for a mobile sidebar navigation drawer.
40. Create a webpage for a dropdown navigation menu.
41. Create a webpage for a mega navigation menu.
42. Create a webpage for a breadcrumb navigation component.
43. Create a webpage for a tab navigation component.
44. Create a webpage for a pill navigation component.
45. Create a webpage for a pagination component.
46. Create a webpage for a SaaS card component.
47. Create a webpage for a profile card component.
48. Create a webpage for a product card component.
49. Create a webpage for a feature card component.
50. Create a webpage for a pricing card component.
51. Create a webpage for a SaaS statistics card component.
52. Create a webpage for a KPI and metric card component.
53. Create a webpage for a dashboard card component.
54. Create a webpage for a notification card component.
55. Create a webpage for an activity feed card component.
56. Create a webpage for a testimonial card component.

57. Create a webpage for a SaaS hero section.
58. Create a webpage for a SaaS landing page header.
59. Create a webpage for primary, secondary, outline, and destructive buttons.
60. Create a webpage for icon buttons and action buttons.
61. Create a webpage for badges, labels, tags, and chips.
62. Create a webpage for user avatars and avatar groups.
63. Create a webpage for dividers and section separators.
64. Create a webpage for alert and notification components.
65. Create a webpage for toast notification components.
66. Create a webpage for tooltip components.
67. Create a webpage for popover components.
68. Create a webpage for modal and dialog components.
69. Create a webpage for confirmation dialog components.
70. Create a webpage for a drawer and slide-out panel.
71. Create a webpage for a command palette interface.
72. Create a webpage for an accordion component.
73. Create a webpage for an empty-state component.
74. Create a webpage for an error-state component.
75. Create a webpage for a success-state component.
76. Create a webpage for a loading-state component.
77. Create a webpage for a skeleton-loading component.
78. Create a webpage for a progress bar component.
79. Create a webpage for a progress stepper component.
80. Create a webpage for a timeline and activity history component.
81. Create a webpage for a login form.
82. Create a webpage for a signup form.
83. Create a webpage for a forgot-password form.
84. Create a webpage for a reset-password form.
85. Create a webpage for a registration form.
86. Create a webpage for a contact form.
87. Create a webpage for a multi-step form.
88. Create a webpage for form validation and error messages.

Modern SaaS, Analytics, AI/ML & Responsive UI

89. Create a webpage for a SaaS dashboard.
90. Create a webpage for an admin dashboard.
91. Create a webpage for an analytics dashboard.
92. Create a webpage for a KPI analytics dashboard.
93. Create a webpage for a revenue analytics dashboard.
94. Create a webpage for a sales analytics dashboard.
95. Create a webpage for user analytics.
96. Create a webpage for customer analytics.
97. Create a webpage for product analytics.
98. Create a webpage for marketing analytics.
99. Create a webpage for financial analytics.
100. Create a webpage for real-time analytics.
101. Create a webpage for SaaS usage analytics.
102. Create a webpage for a conversion funnel dashboard.
103. Create a webpage for a retention and cohort analytics dashboard.

104. Create a webpage for a revenue chart dashboard.
105. Create a webpage for a line chart analytics component.
106. Create a webpage for a bar chart analytics component.
107. Create a webpage for an area chart analytics component.
108. Create a webpage for a pie and donut chart analytics component.
109. Create a webpage for a heatmap analytics component.
110. Create a webpage for data visualization cards.
111. Create a webpage for an analytics date-range filter.
112. Create a webpage for an analytics filter and sorting panel.
113. Create a webpage for customizable dashboard widgets.
114. Create a webpage for a customizable analytics dashboard.
115. Create a webpage for an analytics report and export interface.
116. Create a webpage for an AI assistant interface.
117. Create a webpage for an AI chat interface.
118. Create a webpage for an AI chat sidebar.
119. Create a webpage for an AI command bar.
120. Create a webpage for an AI-powered search interface.
121. Create a webpage for AI search results.
122. Create a webpage for an AI prompt input interface.
123. Create a webpage for AI prompt suggestions.
124. Create a webpage for an AI response card.
125. Create a webpage for an AI-generated content card.
126. Create a webpage for an AI summary card.
127. Create a webpage for an AI recommendation card.
128. Create a webpage for an AI insight card.
129. Create a webpage for an AI copilot interface.
130. Create a webpage for an AI autocomplete interface.
131. Create a webpage for AI smart suggestions.
132. Create a webpage for an AI content generator.
133. Create a webpage for an AI document summarizer.
134. Create a webpage for an AI data analysis interface.
135. Create a webpage for a natural-language query interface.
136. Create a webpage for an Ask Your Data interface.
137. Create a webpage for a text-to-chart analytics interface.
138. Create a webpage for an AI report generator.
139. Create a webpage for an AI-powered analytics dashboard.
140. Create a webpage for an AI prediction card.
141. Create a webpage for an AI forecasting dashboard.
142. Create a webpage for an anomaly detection dashboard.
143. Create a webpage for an AI confidence score component.
144. Create a webpage for an AI sources and citations component.
145. Create a webpage for an AI explanation and reasoning interface.
146. Create a webpage for an AI agent dashboard.
147. Create a webpage for an AI agent list.
148. Create a webpage for an AI agent profile.
149. Create a webpage for AI agent status monitoring.
150. Create a webpage for an AI agent task queue.
151. Create a webpage for an AI agent activity log.

152. Create a webpage for an AI agent execution history.
153. Create a webpage for an AI agent execution timeline.
154. Create a webpage for an AI agent workflow.
155. Create a webpage for an AI agent builder.
156. Create a webpage for AI agent configuration settings.
157. Create a webpage for AI agent permissions management.
158. Create a webpage for an AI agent approval interface.
159. Create a webpage for a human-in-the-loop AI workflow.
160. Create a webpage for an AI agent error and retry interface.
161. Create a webpage for AI agent monitoring.
162. Create a webpage for multi-agent workflow management.
163. Create a webpage for an automation workflow builder.
164. Create a webpage for a trigger and action workflow.
165. Create a webpage for an ML prediction dashboard.
166. Create a webpage for an ML forecast chart.
167. Create a webpage for an ML probability score component.
168. Create a webpage for an ML confidence score component.
169. Create a webpage for an ML risk score dashboard.
170. Create a webpage for a recommendation engine interface.
171. Create a webpage for an ML anomaly alert.
172. Create a webpage for an ML outlier detection interface.
173. Create a webpage for an ML classification result.
174. Create a webpage for an ML prediction comparison dashboard.
175. Create a webpage for an ML model performance dashboard.
176. Create a webpage for ML model metrics.
177. Create a webpage for ML model version management.
178. Create a webpage for ML model monitoring.
179. Create a webpage for ML data drift monitoring.
180. Create a webpage for ML feature importance visualization.
181. Create a webpage for ML prediction explanation.
182. Create a webpage for an ML experiment dashboard.
183. Create a webpage for account settings.
184. Create a webpage for profile settings.
185. Create a webpage for team management.
186. Create a webpage for user management.
187. Create a webpage for role and permission management.
188. Create a webpage for organization settings.
189. Create a webpage for API key management.
190. Create a webpage for API usage analytics.
191. Create a webpage for integrations management.
192. Create a webpage for webhook management.
193. Create a webpage for developer settings.
194. Create a webpage for security settings.
195. Create a webpage for two-factor authentication settings.
196. Create a webpage for active sessions and device management.
197. Create a webpage for an audit log.
198. Create a webpage for SaaS pricing plans.
199. Create a webpage for pricing plan comparison.

200. Create a webpage for subscription management.
201. Create a webpage for subscription upgrade and downgrade.
202. Create a webpage for a SaaS billing dashboard.
203. Create a webpage for payment method management.
204. Create a webpage for invoice management.
205. Create a webpage for invoice details.
206. Create a webpage for usage-based billing.
207. Create a webpage for usage and consumption meters.
208. Create a webpage for AI token and credit usage.
209. Create a webpage for AI spending limits.
210. Create a webpage for billing alerts.
211. Create a webpage for a responsive Flexbox layout.
212. Create a webpage for a responsive CSS Grid layout.
213. Create a webpage for a responsive SaaS dashboard.
214. Create a webpage for a responsive data table.
215. Create a webpage for responsive SaaS cards.
216. Create a webpage for a responsive SaaS form.
217. Create a webpage for a responsive analytics dashboard.
218. Create a webpage for responsive charts and data visualization.
219. Create a webpage for a responsive pricing page.
220. Create a webpage for a responsive login page.
221. Create a webpage for a responsive signup page.
222. Create a webpage for a mobile-first SaaS layout.
223. Create a webpage for a mobile bottom navigation.
224. Create a webpage for a sticky SaaS header.
225. Create a webpage for a fixed SaaS sidebar.
226. Create a webpage for a responsive SaaS footer.
227. Create a webpage for a responsive photo and media gallery.
228. Create a webpage for CSS hover effects and micro-interactions.
229. Create a webpage for animated SaaS cards and components.
230. Create a webpage for a CSS sliding card animation.
231. Create a webpage for a CSS-only modal and popup interface.
232. Create a webpage for a dark-mode SaaS interface using CSS variables.
233. Create a webpage for a light and dark theme switcher.
234. Create a webpage for responsive SaaS typography.
235. Create a webpage for a glassmorphism SaaS interface.
236. Create a webpage for a gradient-based SaaS interface.
237. Create a webpage for a SaaS command palette with keyboard shortcuts.
238. Create a webpage for an AI notification center.
239. Create a webpage for a SaaS user profile menu.
240. Create a webpage for a SaaS onboarding workflow.
241. Create a webpage for a SaaS help and support interface.
242. Create a webpage for a SaaS documentation interface.
243. Create a webpage for a SaaS search and filter interface.
244. Create a webpage for a SaaS calendar and scheduling interface.
245. Create a webpage for a SaaS file management interface.
246. Create a webpage for a SaaS collaboration and team workspace.
247. Final SaaS Webpage Projects

248. Create a webpage for a complete SaaS landing page.
249. Create a webpage for a SaaS login page.
250. Create a webpage for a SaaS signup page.
251. Create a webpage for a SaaS onboarding page.
252. Create a webpage for a complete SaaS dashboard.
253. Create a webpage for a complete analytics dashboard.
254. Create a webpage for an AI-powered SaaS dashboard.
255. Create a webpage for an AI agent management dashboard.
256. Create a webpage for a SaaS user management system.
257. Create a webpage for a SaaS team management system.
258. Create a webpage for a SaaS account settings page.
259. Create a webpage for a SaaS security settings page.
260. Create a webpage for a SaaS integrations page.
261. Create a webpage for a SaaS billing management page.
262. Create a webpage for a SaaS pricing page.
263. Create a webpage for a SaaS checkout page.
264. Create a webpage for a SaaS invoice management page.
265. Create a webpage for a SaaS help center.
266. Create a webpage for a SaaS documentation website.
267. Create a webpage for a SaaS blog and news website.
268. Create a webpage for a SaaS 404 error page.
269. Create a webpage for a SaaS 500 error page.
270. Create a webpage for a complete responsive SaaS website using HTML5, CSS3, Flexbox, Grid, media queries, analytics, AI/ML interfaces, dashboards, and modern SaaS components.

JavaScript

1. Write a JavaScript program to display "Hello World".
2. Write a JavaScript program to perform arithmetic operations.
3. Write a program to find the largest of three numbers.
4. Write a program to check whether a number is even or odd.
5. Write a program to check whether a number is positive, negative, or zero.
6. Write a program to find the factorial of a number.
7. Write a program to generate the Fibonacci series.
8. Write a program to check whether a number is prime.
9. Write a program to check whether a number is a palindrome.
10. Write a program to reverse a number.
11. Write a program to find the sum of digits of a number.
12. Write a program to find the largest and smallest element in an array.
13. Write a program to sort an array.
14. Write a program to find the sum and average of array elements.
15. Write a program to demonstrate JavaScript functions.
16. Write a program using arrow functions.
17. Write a program demonstrating strings and string methods.
18. Write a program demonstrating arrays and array methods.
19. Write a program demonstrating objects in JavaScript.
20. Write a program demonstrating Date and Math objects.

21. DOM & Event-Based Experiments
22. Create a webpage that changes text using JavaScript.
23. Change the background color of a webpage using JavaScript.
24. Create a button click event using JavaScript.
25. Create a program to show/hide an HTML element.
26. Create a program to change an image when a button is clicked.
27. Create a digital clock using JavaScript.
28. Create a simple calculator using HTML, CSS, and JavaScript.
29. Create a number guessing game.
30. Create a form validation program.
31. Validate name, email, phone number, and password using JavaScript.
32. Create a to-do list using JavaScript.
33. Create a simple quiz application.
34. Create a counter application with increment and decrement buttons.
35. Create a stopwatch using JavaScript.
36. Create a countdown timer using JavaScript.
37. Advanced JavaScript Lab Questions
38. Demonstrate ES6 let, const, template literals, and destructuring.
39. Demonstrate spread and rest operators.
40. Demonstrate classes and objects.
41. Demonstrate inheritance in JavaScript.
42. Demonstrate callbacks, promises, and async/await.
43. Fetch data from an API using Fetch API.
44. Display API data dynamically on a webpage.
45. Store and retrieve data using localStorage.
46. Create a responsive login form with JavaScript validation.
47. Create a complete interactive webpage using HTML, CSS, and JavaScript.
48. Write a JavaScript program to swap two numbers.
49. Write a program to find the greatest common divisor (GCD) of two numbers.
50. Write a program to find the least common multiple (LCM) of two numbers.
51. Write a program to check whether a number is an Armstrong number.
52. Write a program to check whether a number is a perfect number.
53. Write a program to generate the multiplication table of a given number.
54. Write a program to calculate the power of a number.
55. Write a program to count the number of digits in a number.
56. Write a program to find the second-largest element in an array.
57. Write a program to remove duplicate elements from an array.
58. Write a program to merge two arrays.
59. Write a program to find the frequency of each element in an array.
60. Write a program to find common elements in two arrays.
61. Write a program to reverse a string.
62. Write a program to check whether a string is a palindrome.
63. Write a program to count vowels and consonants in a string.
64. Write a program to count the number of words in a string.
65. Write a program to find the frequency of characters in a string.
66. Write a program to demonstrate regular expressions in JavaScript.
67. Write a program to validate an email address using a regular expression.

DOM Manipulation

1. Create a webpage and use JavaScript to add new elements dynamically.
2. Create a program to remove HTML elements dynamically.
3. Create a program to change CSS styles using JavaScript.
4. Create a program to change multiple HTML elements simultaneously.
5. Demonstrate getElementById() and getElementsByClassName().
6. Demonstrate querySelector() and querySelectorAll().
7. Create a program to create and remove CSS classes dynamically.
8. Create a dynamic table using JavaScript.
9. Create a program to add and delete rows from a table.
10. Create a program to search/filter table records.
11. Create a live character counter for a textarea.
12. Create a password show/hide feature.
13. Create a dynamic dropdown list.
14. Create a dependent dropdown such as Country → State → City.
15. Create an image slideshow/carousel using JavaScript.

Events

1. Demonstrate mouse events using JavaScript.
2. Demonstrate keyboard events using JavaScript.
3. Demonstrate form events such as submit, change, focus, and blur.
4. Create a webpage that responds to double-click events.
5. Create a program using event bubbling and event capturing.
6. Demonstrate event delegation.
7. Create a keyboard-controlled webpage/game.
8. Create a drag-and-drop application using JavaScript.

Forms & Validation

1. Create a student registration form with JavaScript validation.
2. Create a login form with username and password validation.
3. Create a password strength checker.
4. Create a signup form with password confirmation.
5. Create a feedback form with validation.
6. Create a college admission form with JavaScript validation.
7. Display validation error messages dynamically without reloading the page.

Browser & Storage

1. Demonstrate the Window object.
2. Demonstrate the Navigator object.
3. Demonstrate the Location object.
4. Demonstrate the History object.
5. Create a program using localStorage.
6. Create a program using sessionStorage.
7. Create a shopping cart using localStorage.
8. Create a theme switcher (light/dark mode) using localStorage.

Mini Projects

1. Create a digital calculator.

2. Create a digital clock.
3. Create a stopwatch.
4. Create a countdown timer.
5. Create a to-do list application.
6. Create a weather application using an API.
7. Create a quiz application.
8. Create a number guessing game.
9. Create a tic-tac-toe game.
10. Create a simple expense tracker.
11. Create a student marks/grade calculator.
12. Create a simple shopping cart.
13. Create a notes application using localStorage.
14. Create a random password generator.
15. Create a BMI calculator.
16. Create a currency converter using an API.
17. Create a complete interactive portfolio website using HTML, CSS, and JavaScript.

DOM Manipulation

1. Change the text content of an HTML element using JavaScript.
2. Change the HTML content of an element using innerHTML.
3. Change the style of an HTML element dynamically.
4. Change the background color of an element when a button is clicked.
5. Create a new HTML element dynamically and add it to the webpage.
6. Remove an existing HTML element dynamically.
7. Replace an existing HTML element with a new element.
8. Add and remove CSS classes using JavaScript.
9. Toggle a CSS class using JavaScript.
10. Change the attributes of an HTML element dynamically.
11. Create a dynamic unordered list from an array.
12. Create a dynamic ordered list from user input.
13. Create a dynamic student table using JavaScript.
14. Add new rows to an HTML table dynamically.
15. Delete a selected row from an HTML table.
16. Edit table data dynamically.
17. Create a searchable/filterable table.
18. Sort table records using JavaScript.
19. Create a dynamic dropdown menu using JavaScript.
20. Add options dynamically to a <select> element.
21. Create a dependent dropdown — Country → State → City.
22. Create an image gallery dynamically from an array of image data.
23. Create an image slideshow using DOM manipulation.
24. Create a show/hide password feature.
25. Create a show/hide content button.
26. Create a tabs interface using DOM manipulation.
27. Create an accordion using JavaScript.
28. Create a modal popup using JavaScript.
29. Create a tooltip dynamically.
30. Create a live character counter for a textarea.

31. Create a live word counter.
32. Create a program to dynamically display the current date and time.
33. Create a to-do list using DOM manipulation.
34. Create a program to add, edit, and delete to-do items.
35. Demonstrate createElement(), appendChild(), remove(), and replaceChild().

2. JavaScript Events

1. Demonstrate the onclick event.
2. Demonstrate the ondblclick event.
3. Demonstrate onmouseover and onmouseout.
4. Demonstrate onmousedown and onmouseup.
5. Demonstrate the onmousemove event.
6. Demonstrate keyboard events using keydown.
7. Demonstrate keyboard events using keyup.
8. Create a webpage that displays the key pressed by the user.
9. Create a program that counts the number of keyboard key presses.
10. Demonstrate the focus and blur events.
11. Demonstrate the change event.
12. Demonstrate the input event.
13. Demonstrate the submit event.
14. Demonstrate the reset event.
15. Create a button that changes the background color when clicked.
16. Create a button that changes font size dynamically.
17. Create a webpage with multiple buttons having different event handlers.
18. Demonstrate event listeners using addEventListener().
19. Add multiple event listeners to a single element.
20. Remove an event listener using removeEventListener().
21. Demonstrate event bubbling.
22. Demonstrate event capturing.
23. Demonstrate event delegation.
24. Create a drag-and-drop application.
25. Create a mouse coordinate tracker.
26. Create a keyboard-controlled moving object.
27. Create a click counter.
28. Create a double-click counter.
29. Create a reaction-time game.
30. Create a simple keyboard typing test.

3. Forms and Validation

1. Create a student registration form and validate all fields.
2. Create a login form with username and password validation.
3. Validate a name field to accept only alphabets.
4. Validate an age field using JavaScript.
5. Validate a mobile phone number.
6. Validate an email address.
7. Validate a password based on minimum length.
8. Create a strong password validator.
9. Validate password and confirm password fields.

10. Validate a date of birth field.
11. Validate a gender selection.
12. Validate radio buttons.
13. Validate checkbox selection.
14. Validate a dropdown/select box.
15. Validate a textarea.
16. Create a form with required-field validation.
17. Display error messages beside individual form fields.
18. Display a success message after successful form submission.
19. Prevent form submission when validation fails.
20. Validate a form using regular expressions.
21. Validate an email using a regular expression.
22. Validate a phone number using a regular expression.
23. Validate a password using a regular expression.
24. Create a real-time form validation system.
25. Validate fields using the input, change, focus, and blur events.
26. Create a password strength meter.
27. Create a show/hide password option.
28. Create a registration form with password confirmation.
29. Create a college admission form with complete validation.
30. Create a job application form with validation.
31. Create a feedback form with validation.
32. Create a contact form with validation.
33. Create a multi-step registration form.
34. Create a form that displays a preview of entered information before submission.
35. Create a form that calculates and displays age from date of birth.
36. Create a marks entry form and calculate total, average, and grade.
37. Create an employee salary form and calculate salary details.
38. Create a shopping order form and calculate the total price.
39. Create a form with dynamic fields based on user selection.
40. Create a form that validates input and displays errors without page reload.

4. Browser Objects & Browser Features

1. Demonstrate the Window object.
2. Demonstrate alert(), confirm(), and prompt().
3. Create a program using setTimeout().
4. Create a program using setInterval().
5. Create a digital clock using setInterval().
6. Create a countdown timer.
7. Create a stopwatch.
8. Demonstrate the Location object.
9. Create a button to redirect the user to another webpage.
10. Display the current webpage URL using JavaScript.
11. Demonstrate the History object.
12. Create buttons for Back, Forward, and Reload browser operations.
13. Demonstrate the Navigator object.
14. Display browser information using JavaScript.
15. Display the user's screen width and height.
16. Detect whether the browser is online or offline.

17. Display the current browser language.
18. Demonstrate browser geolocation.
19. Display the user's current geographical coordinates using the Geolocation API.
20. Create a webpage that updates information when the browser goes online/offline.

5. Web Storage

1. Store a username using localStorage.
2. Retrieve and display data from localStorage.
3. Update data stored in localStorage.
4. Delete an item from localStorage.
5. Clear all localStorage data.
6. Store multiple values using localStorage.
7. Store an array in localStorage using JSON.
8. Store a JavaScript object in localStorage.
9. Create a student record system using localStorage.
10. Create a to-do list using localStorage.
11. Create a notes application using localStorage.
12. Create a shopping cart using localStorage.
13. Create a login preference system using localStorage.
14. Create a dark/light theme switcher using localStorage.
15. Save the user's preferred font size using localStorage.
16. Create a recently viewed items feature using localStorage.
17. Create a visitor counter using browser storage.
18. Compare localStorage and sessionStorage.
19. Store temporary form data using sessionStorage.
20. Create a multi-page form using sessionStorage.
21. Save form data automatically using localStorage.
22. Restore previously entered form data after refreshing the page.
23. Create a shopping cart that persists after page refresh.
24. Create a student marks database using localStorage.
25. Create a CRUD application using localStorage — Create, Read, Update, and Delete records.

MYSQL - Database

1. Database Fundamentals
2. Create a database using MySQL.
3. Create and delete tables using MySQL.
4. Demonstrate different MySQL data types.
5. Demonstrate primary key and foreign key.
6. Implement NOT NULL, UNIQUE, DEFAULT, and CHECK constraints.
7. Alter table structure using ALTER TABLE.
8. Rename and truncate tables.

SQL Commands

1. Insert records using INSERT.
2. Retrieve records using SELECT.

3. Update records using UPDATE.
4. Delete records using DELETE.
5. Use WHERE clause for filtering records.
6. Use ORDER BY for sorting records.
7. Use DISTINCT to retrieve unique values.
8. Use LIMIT to restrict query results.

SQL Operators and Functions

1. Demonstrate arithmetic operators in SQL.
2. Demonstrate comparison and logical operators.
3. Use LIKE, IN, and BETWEEN operators.
4. Demonstrate aggregate functions: COUNT(), SUM(), AVG(), MIN(), and MAX().
5. Demonstrate string functions.
6. Demonstrate date and time functions.
7. Demonstrate GROUP BY and HAVING.

Joins

1. Implement INNER JOIN.
2. Implement LEFT JOIN.
3. Implement RIGHT JOIN.
4. Implement self join.
5. Implement multiple-table joins.

Subqueries

1. Implement single-row subqueries.
2. Implement multiple-row subqueries.
3. Implement subqueries with IN.
4. Implement correlated subqueries.

Views and Indexes

1. Create and use views.
2. Update and delete views.
3. Create indexes on tables.
4. Demonstrate unique indexes.
5. Drop indexes.

Stored Procedures and Functions

1. Create and execute a stored procedure.
2. Create a stored procedure with parameters.
3. Create a stored function.
4. Use conditional statements inside stored procedures.
5. Use loops inside stored procedures.

Triggers

1. Create a BEFORE INSERT trigger.
2. Create an AFTER INSERT trigger.
3. Create an BEFORE UPDATE trigger.
4. Create an AFTER DELETE trigger.
5. Implement an audit table using triggers.

Transactions

1. Demonstrate transactions in MySQL.
2. Implement COMMIT.
3. Implement ROLLBACK.
4. Demonstrate SAVEPOINT.

5. Demonstrate transaction handling with multiple SQL operations.

Database Design

1. Design a Student Management Database.
2. Design a User Management Database.
3. Design a Product Management Database.
4. Design a Library Management Database.
5. Implement relationships between multiple tables.
6. Implement normalization using 1NF, 2NF, and 3NF.

CRUD Operations

1. Implement CRUD operations using SQL.
2. Create a Student Management CRUD database.
3. Create a User Management CRUD database.
4. Create a Product Management CRUD database.
5. Implement CRUD operations using multiple related tables.
6. Implement input constraints for CRUD operations.

Advanced SQL

1. Implement UNION and UNION ALL.
2. Implement CASE statements.
3. Implement Common Table Expressions (CTEs).
4. Demonstrate window functions.
5. Implement EXISTS and NOT EXISTS.
6. Demonstrate ON DELETE CASCADE and ON UPDATE CASCADE.
7. Export and import a MySQL database.
8. Backup and restore a MySQL database.

Node.js Backend — Experiment List

Server-Side Architecture

1. Implement a basic Client-Server Architecture using Node.js.
2. Create a basic HTTP Server using Node.js.
3. Demonstrate the Request-Response Cycle in a server-side application.
4. Node.js Fundamentals
5. Create a Node.js application using built-in modules.
6. Demonstrate the File System (fs) module.
7. Demonstrate the Path (path) module.
8. Create and use custom Node.js modules.
9. Install and use packages using npm.
10. Demonstrate Callbacks, Promises, and Async/Await in Node.js.
11. Implement Error Handling in a Node.js application.

Express Framework

1. Create a basic Express.js application.
2. Handle GET and POST requests using Express.
3. Serve static files using Express.
4. Send HTML and JSON responses using Express.
5. Create an Express application with a modular project structure.

Routing

1. Implement basic routing using Express.
2. Implement route parameters and query parameters.

3. Create multiple routes using Express Router.
4. Implement 404 / Not Found route handling.
5. Create modular routes for users, products, or students.

Middleware

1. Create and implement custom middleware.
2. Implement logging middleware.
3. Implement request validation middleware.
4. Implement authentication middleware.
5. Implement error-handling middleware.
6. Demonstrate the execution order of middleware.

Session and Cookie Management

1. Create and manage cookies using Express.
2. Read and delete cookies using Express.
3. Implement session management in Express.
4. Implement login and logout using sessions.
5. Create protected routes using session authentication.

Database Connectivity

1. Connect a Node.js application to a database.
2. Create and manage database tables/collections.
3. Insert data into a database using Node.js.
4. Retrieve data from a database using Node.js.
5. Update database records using Node.js.
6. Delete database records using Node.js.

CRUD Operations

1. Implement CRUD operations using Node.js and Express.
2. Create a Student Management CRUD application.
3. Create a User Management CRUD application.
4. Create a Product Management CRUD application.
5. Implement CRUD operations with database connectivity.
6. Implement input validation for CRUD operations.

Authentication

1. Implement User Registration.
2. Implement User Login and Logout.
3. Implement Password Hashing.
4. Implement Password Verification.
5. Implement JWT-based Authentication.
6. Create Protected API Routes using authentication.

Authorization

1. Implement Role-Based Authorization.
2. Create Admin and User roles.
3. Restrict specific routes based on user roles.
4. Implement Authorization Middleware.
5. RESTful API Development
6. Create a basic RESTful API using Express.js.
7. Implement a GET REST API.
8. Implement a POST REST API.
9. Implement a PUT/PATCH REST API.
10. Implement a DELETE REST API.

11. Implement proper HTTP Status Codes in REST APIs.
12. Implement API Error Handling.
13. Implement API Request Validation.
14. Test REST APIs using Postman.
15. Create a Student REST API with Database Connectivity.
16. Create a Product REST API with CRUD Operations.
17. Create a User REST API with Authentication and Authorization.

SUPABASE

Supabase Fundamentals

1. Create a Supabase project.
2. Explore the Supabase Dashboard.
3. Configure project settings and API keys.
4. Create and manage a PostgreSQL database using Supabase.
5. Demonstrate Supabase Table Editor.
6. Demonstrate Supabase SQL Editor.
7. Connect an application to Supabase.

Database and Tables

1. Create tables using Supabase.
2. Define columns and PostgreSQL data types.
3. Implement primary keys and foreign keys.
4. Implement NOT NULL, UNIQUE, DEFAULT, and CHECK constraints.
5. Insert records into Supabase tables.
6. Retrieve records from Supabase tables.
7. Update records in Supabase tables.
8. Delete records from Supabase tables.

SQL Operations

1. Execute basic SELECT queries using Supabase SQL Editor.
2. Implement filtering using WHERE.
3. Implement sorting using ORDER BY.
4. Implement GROUP BY and HAVING.
5. Use aggregate functions such as COUNT, SUM, AVG, MIN, and MAX.
6. Implement SQL joins.
7. Implement subqueries.
8. Create and use database views.

Supabase CRUD

1. Implement Create operations using Supabase Client.
2. Implement Read operations using Supabase Client.
3. Implement Update operations using Supabase Client.
4. Implement Delete operations using Supabase Client.
5. Create a Student Management CRUD application.
6. Create a Product Management CRUD application.
7. Create a User Management CRUD application.

Authentication

1. Configure Supabase Authentication.

2. Implement user registration.
3. Implement user login.
4. Implement user logout.
5. Retrieve the currently authenticated user.
6. Implement email/password authentication.
7. Implement password reset.
8. Manage authentication sessions.
9. Implement protected pages/routes using Supabase Authentication.

Authorization and Security

1. Enable Row Level Security (RLS).
2. Create RLS policies.
3. Restrict users to their own records.
4. Create Admin and User roles.
5. Implement role-based access control.
6. Implement authenticated-user policies.
7. Test database access with different users.

Storage

1. Create a Supabase Storage bucket.
2. Upload files using Supabase Storage.
3. Retrieve files from Storage.
4. Download files from Storage.
5. Delete files from Storage.
6. Generate public file URLs.
7. Implement image/file upload with authentication.

Realtime

1. Enable Supabase Realtime.
2. Listen for database INSERT events.
3. Listen for database UPDATE events.
4. Listen for database DELETE events.
5. Create a realtime chat application.
6. Create a realtime student record application.

Edge Functions

1. Create a Supabase Edge Function.
2. Deploy an Edge Function.
3. Pass data to an Edge Function.
4. Return JSON responses from an Edge Function.
5. Implement authentication inside an Edge Function.
6. Connect an Edge Function with a Supabase database.

REST API

1. Access Supabase tables through the generated REST API.
2. Implement GET operations using Supabase REST API.
3. Implement POST operations using Supabase REST API.
4. Implement PATCH/UPDATE operations using Supabase REST API.
5. Implement DELETE operations using Supabase REST API.
6. Implement API filtering and pagination.
7. Test Supabase REST APIs using Postman.

Project-Based Experiments

1. Create a Student Management System using Supabase.

2. Create a User Authentication System using Supabase.
3. Create a Product Management System using Supabase.
4. Create a Realtime Chat Application using Supabase.
5. Create a File Upload Application using Supabase Storage.
6. Create a Student REST API using Supabase.
7. Create a Role-Based Admin Dashboard using Supabase Authentication and RLS.